Henry — 39/40 (Q1 11/12 · Q2 13/13 · Q3 15/15)

南京外国语学校
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# CIE Semester 1 Midterm Examination 2026 - 2027

## Paper 1&2 Combined

## AS Computer science

**60 Minutes**

No additional materials are needed

**READ THESE INSTRUCTIONS FIRST**

Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use red ink, staples, paper clips, highlighters, glue or correction fluid.
**The exam paper contains 3 structured questions.**
Answer **all** your questions on the **exam booklet.**

| | |
|---|---|
| **Q1** | **/ 12** |
| **Q2** | **/ 13** |
| **Q3** | **/ 15** |
| **Total** | **/ 40** |

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of 8 printed pages.

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**[TURN OVER]**

**1.** A gym uses a program to store information about its members.

(a) State the most appropriate **data type** for each item of data.

| Item of data | Data type |
|---|---|
| the membership number — a whole number between 1 and 9999 | INTEGER |
| the monthly fee — for example 299.50 | REAL |
| whether the member has paid for this month | BOOLEAN |
| the membership grade — a single letter, A, B, C or D | CHAR |

[4]

(b) The following pseudocode is run.

```
DECLARE Count : INTEGER
DECLARE Total : INTEGER

Count <- 1
Total <- 0

WHILE Count <= 4
   Total <- Total + Count * 2
   Count <- Count + 1
ENDWHILE

OUTPUT Total
OUTPUT Count
```

Complete the trace table below.

| Count | Count <= 4 ? | Total |
|---|---|---|
| 1 | TRUE | 2 |
| 2 | TRUE | 6 |
| 3 | TRUE | 12 |
| 4 | TRUE | 20 |
| 5 | FALSE | 20 |

Output:

Total = 20

Count = 5 [4]

(c) A programmer writes the following pseudocode. It is intended to output the numbers 10, 9, 8, 7, 6, 5, 4, 3, 2, 1.

```
DECLARE N : INTEGER

N <- 10

WHILE N >= 0
   OUTPUT N
   N <- N + 1
ENDWHILE
```

(i) State the error in this pseudocode and explain what happens when it is run.

runtime error

The while loop never ends when it is run as N is always larger than 0

[2]

(ii) Rewrite the pseudocode so that it outputs the numbers as intended.

```
DECLARE N: INTEGER
N ← 10
WHILE N > 0
   OUTPUT N
   N ← N - 1
ENDWHILE
```

[2]

Q1(c)(i) 1/2

You found the infinite loop (N keeps growing, so the test never becomes false). The second error is the condition WHILE N >= 0: counting down, 0 would also be printed and the intended output stops at 1, so it must be N > 0. Both errors must be identified for the 2 marks.

**2.** The following pseudocode defines a function.

```
FUNCTION Calc(A : INTEGER, B : INTEGER) RETURNS INTEGER
   IF A > B THEN
      RETURN A * 2
   ELSE
      RETURN B + A
   ENDIF
ENDFUNCTION
```

(a) State the value returned by each of the following calls.

| Call | Value returned |
|---|---|
| `Calc(7, 3)` | 14 |
| `Calc(2, 9)` | 11 |
| `Calc(5, 5)` | 10 |

[3]

(b) Write pseudocode for a function `DigitTotal` that takes an `INTEGER` parameter and **returns** the sum of its digits. For example, `DigitTotal(472)` returns 13.

```
FUNCTION DigitTotal (num: INTEGER) RETURNS INTE
DECLARE sum, digit, i : INTEGER
sum ← 0 DECLARE str : STRING
digit ←0 FOR i ← 1 TO Len(NUM_TO_STR
sum ←0 digit ← STR_TO_NUM(MID(
str ← NUM_TO_STR(num)
FOR i ← 1 TO Len(str)
    digit ← STR_TO_NUM(MID(str, i, 1))
    sum ← sum + digit
Next i
Return sum
ENDFUNCTION
```

[5]

(c) Explain the difference between a **function** and a **procedure**.

a function returns a value, while a procedure doesn't.

[2]

(d) Explain what is meant by **modularization** and give **one** benefit of using it.

modularization means changing code into the form of function or procedure.
It makes code easier to maintain and more readable.

[3]

**3.** A school stores the scores of 30 students in a one-dimensional array `Scores`.

(a) Write pseudocode to declare the array `Scores`.

```
DECLARE Scores : ARRAY [1:30] OF INTEGER
```

[2]

(b) The array `Scores` has been filled with 30 values. Write pseudocode to output the **largest** value in `Scores` and the **index** at which it is found.

```
DECLARE largest,index,i : INTEGER
largest ← Scores [1]
FOR i ← 2 TO 30
    IF Scores [i] > largest THEN
        largest ← Scores [i]
        index ← i
    ENDIF
NEXT i
OUTPUT largest
OUTPUT index
```

[5]

(c) Write pseudocode for a function `CountPass` that takes the array `Scores` and an integer `PassMark` as parameters and **returns** the number of values in `Scores` that are **greater than or equal to** `PassMark`.

```
FUNCTION CountPass (Scores : ARRAY, PassMark : INTEGER) RETURNS INTEGER
    DECLARE i, count : INTEGER
    count ← 0
    FOR i ← 1 TO 30
        IF Scores[i] >= PassMark THEN
            count ← count + 1
        ENDIF
    NEXT i
    RETURN count
ENDFUNCTION
```

[5]

(d) The school also stores the marks for three subjects in a two-dimensional array `Marks`. `Marks` has **5 rows** and **3 columns**.

(i) Write pseudocode to declare the array `Marks`.

```
DECLARE Marks : ARRAY [1:5, 1:3] OF INTEGER
```

[1]

(ii) Write pseudocode to output **every** value in `Marks`.

```
DECLARE i, j : INTEGER
FOR i ← 1 TO 5
    FOR j ← 1 TO 3
        OUTPUT (Marks[i, j])
    NEXT j
NEXT i
```

[2]

End of the exam.

Use following pages for scratch paper.